Jayant Opencast Project (OCP)

🏭 Introduction

The **Jayant Opencast Project (OCP)** is one of the largest and most productive coal mines operated by Northern Coalfields Limited, a subsidiary of Coal India Limited. Located in the Singrauli coalfield, it is known for its high-capacity production and advanced mechanization.

Commissioned in the mid-1980s, Jayant OCP has grown into a **mega opencast mine** with an annual production capacity of around **25–30 million tonnes**, making it one of the highest-producing mines in the NCL command area. The mine plays a crucial role in supplying coal to major thermal power plants, especially those operated by NTPC Limited.

The project uses large-scale mining equipment such as **draglines, electric shovels, and high-capacity dumpers**, and is often studied as a model for **high-efficiency surface mining operations in India**.

🌍 Geographical Overview – Jayant Opencast Project (OCP)

The Jayant OCP is located in the **Singrauli coalfield**, primarily in the Singrauli district of Madhya Pradesh, close to the border of Uttar Pradesh. This region is one of the most significant coal-producing belts in India.

📍 Location & Setting

- Situated in the **central part of the Singrauli coalfield**
- Lies within the **Moher sub-basin**, a major coal-bearing zone
- Well connected via rail and road for efficient coal dispatch

🌄 Physical & Geological Features

- Located in the **Vindhyan plateau region**, featuring gently undulating terrain
- Contains **thick and laterally extensive coal seams**, ideal for opencast mining
- Coal type: **non-coking, sub-bituminous coal**, mainly used for power generation

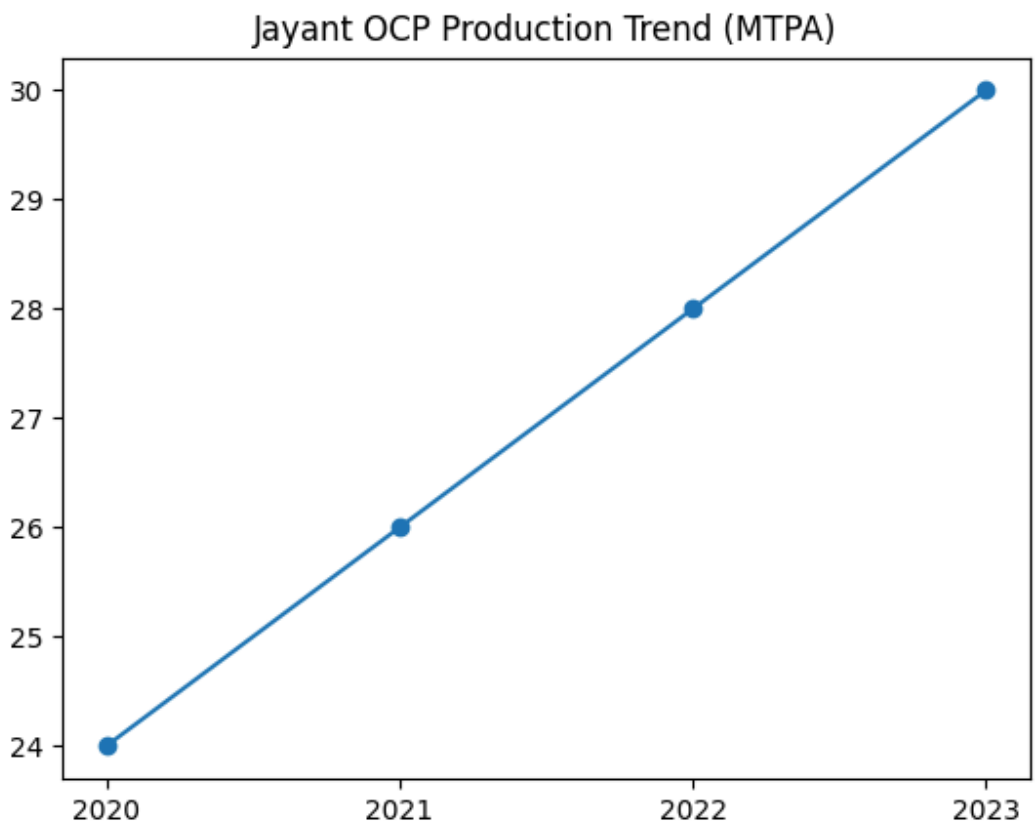
🌊 Nearby Natural Features

- Close to the Rihand River and its reservoir
- The region supports several thermal power stations due to proximity to water and coal

🌦️ Climate

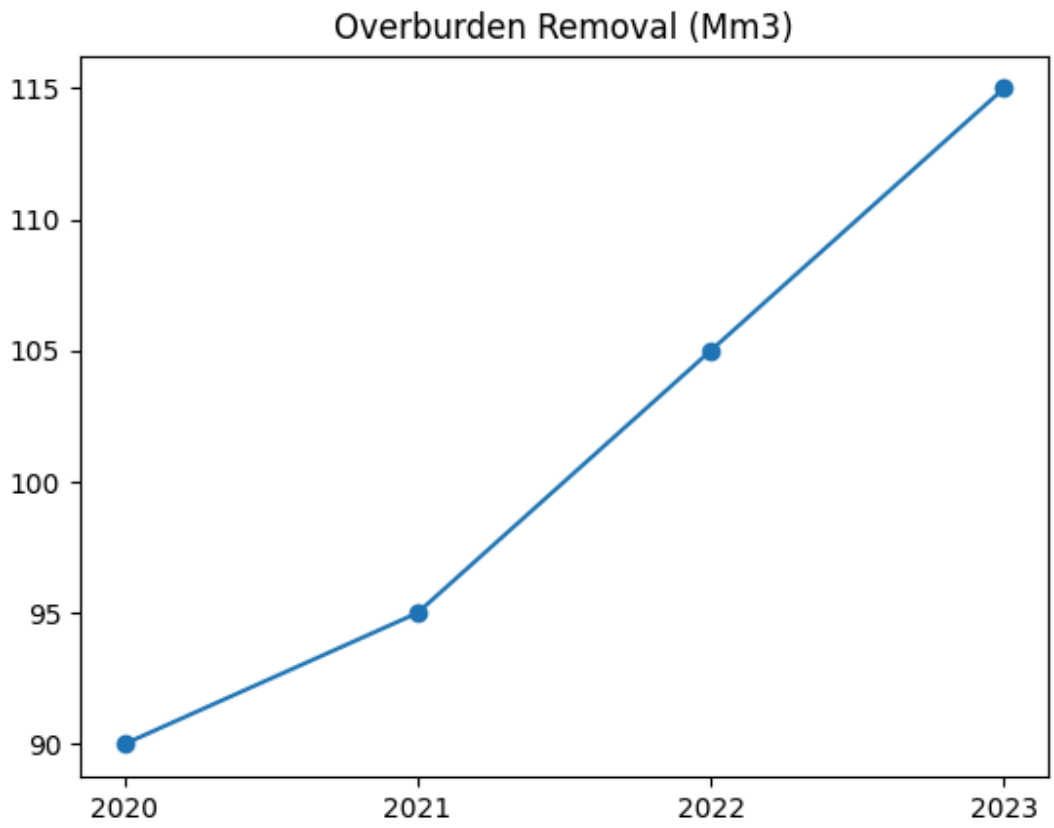
- Tropical climate with extreme seasonal variation
- Summers: very hot (often above 45°C)
- Monsoon: moderate to heavy rainfall
- Winters: mild to cool

📊 1. Production Trend (MTPA)



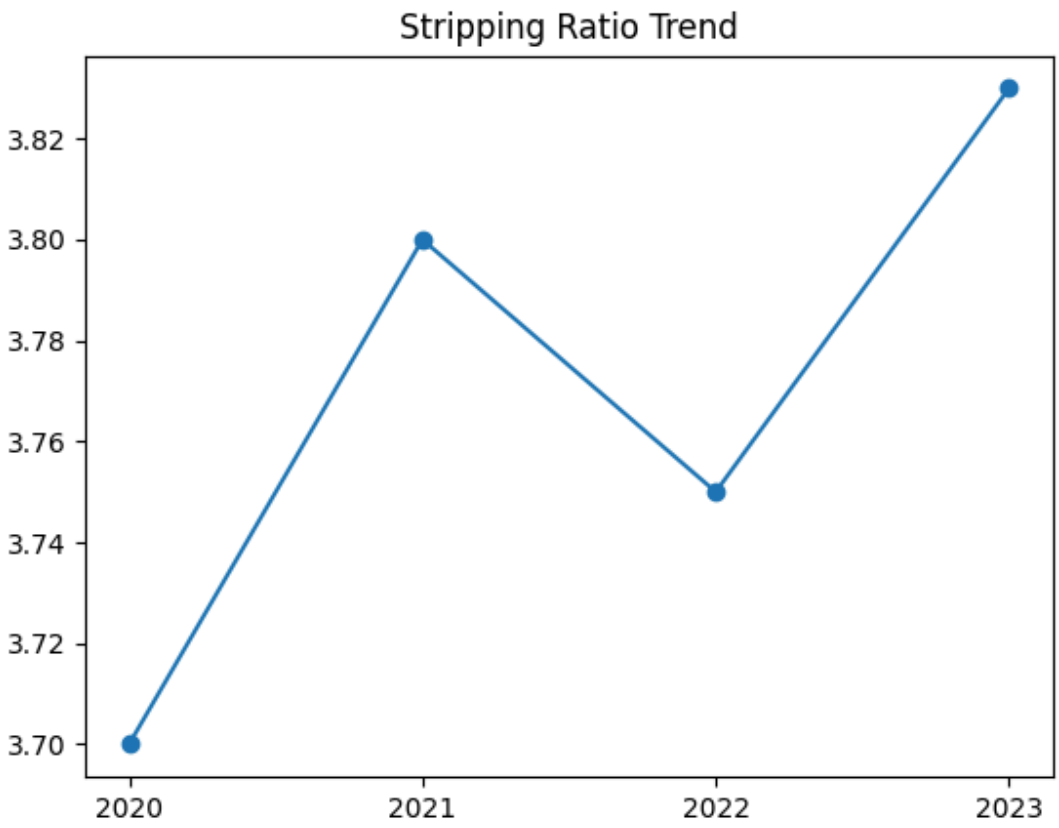
This chart shows the year-wise increase in coal production at Jayant OCP, operated by Northern Coalfields Limited.
It reflects consistent growth due to mechanization and expansion.
Significance: Helps in production forecasting, capacity planning, and evaluating mine performance.

📊 2. Overburden Removal (Mm³)



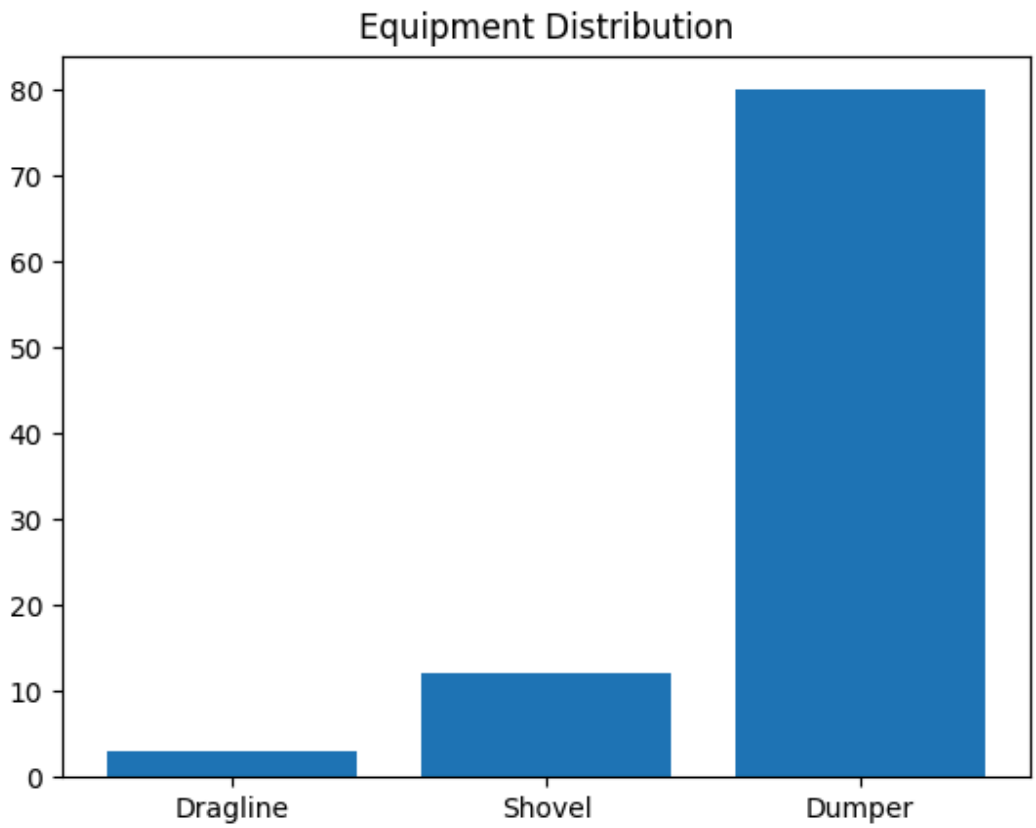
This graph represents the quantity of overburden removed over time.
An increasing trend indicates deeper mining operations.
Significance: Critical for estimating stripping ratio, dump design, and equipment requirement.

📊 3. Stripping Ratio Trend



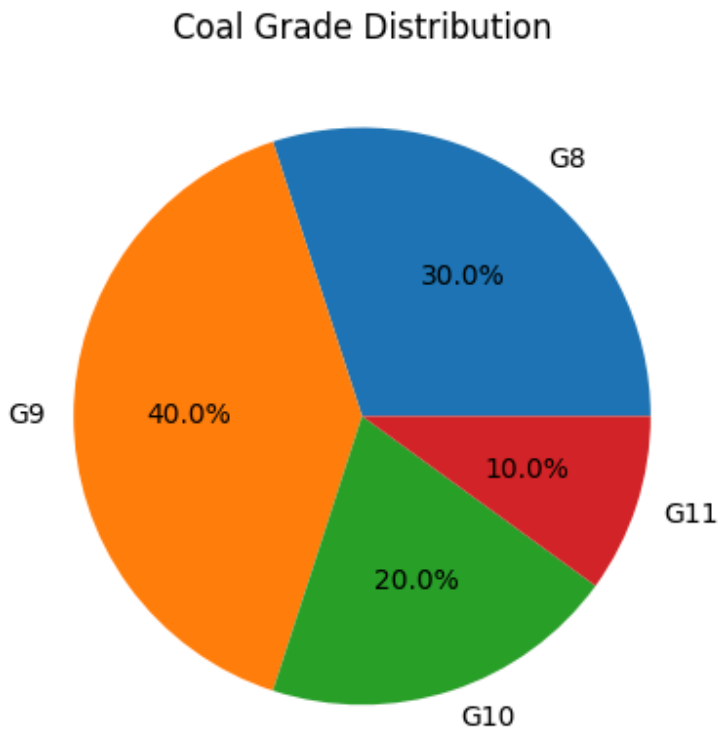
Shows variation in stripping ratio across years.
Stable values indicate efficient mine planning and controlled cost.
Significance: One of the most important economic indicators in opencast mining.

📊 4. Equipment Distribution



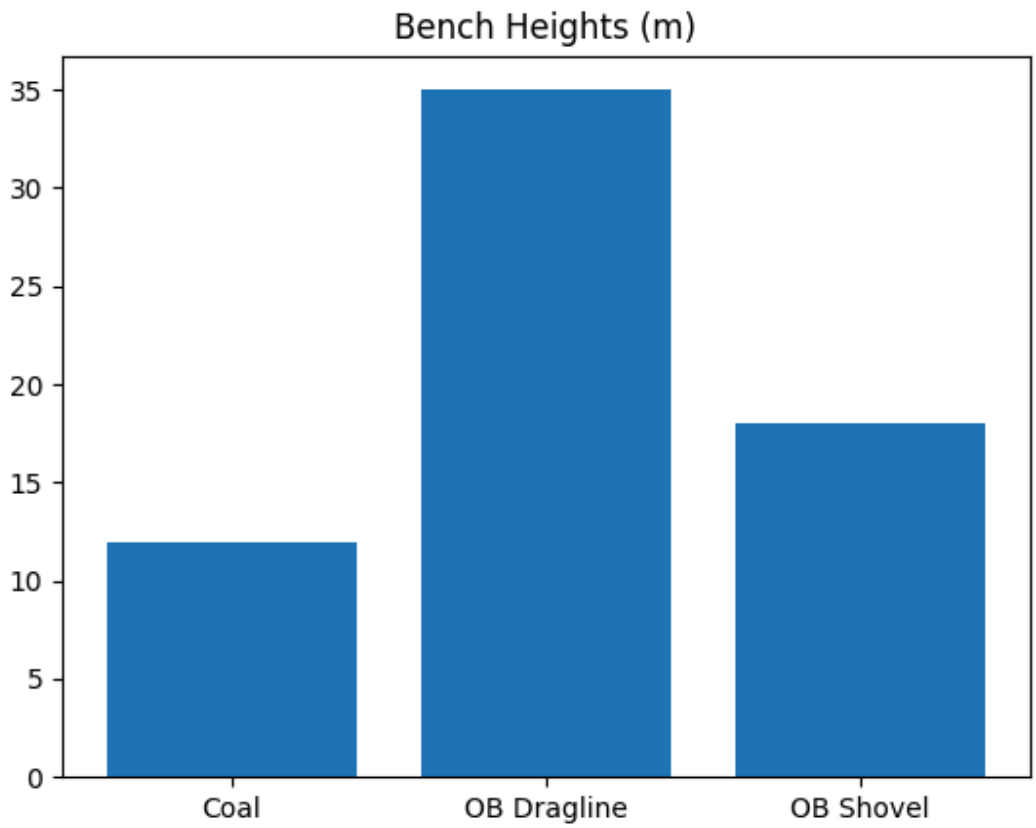
Displays allocation of major mining equipment such as draglines, shovels, and dumpers. Dumpers dominate due to large-scale material handling.
Significance: Helps in equipment optimization and productivity analysis.

📊 5. Coal Grade Distribution



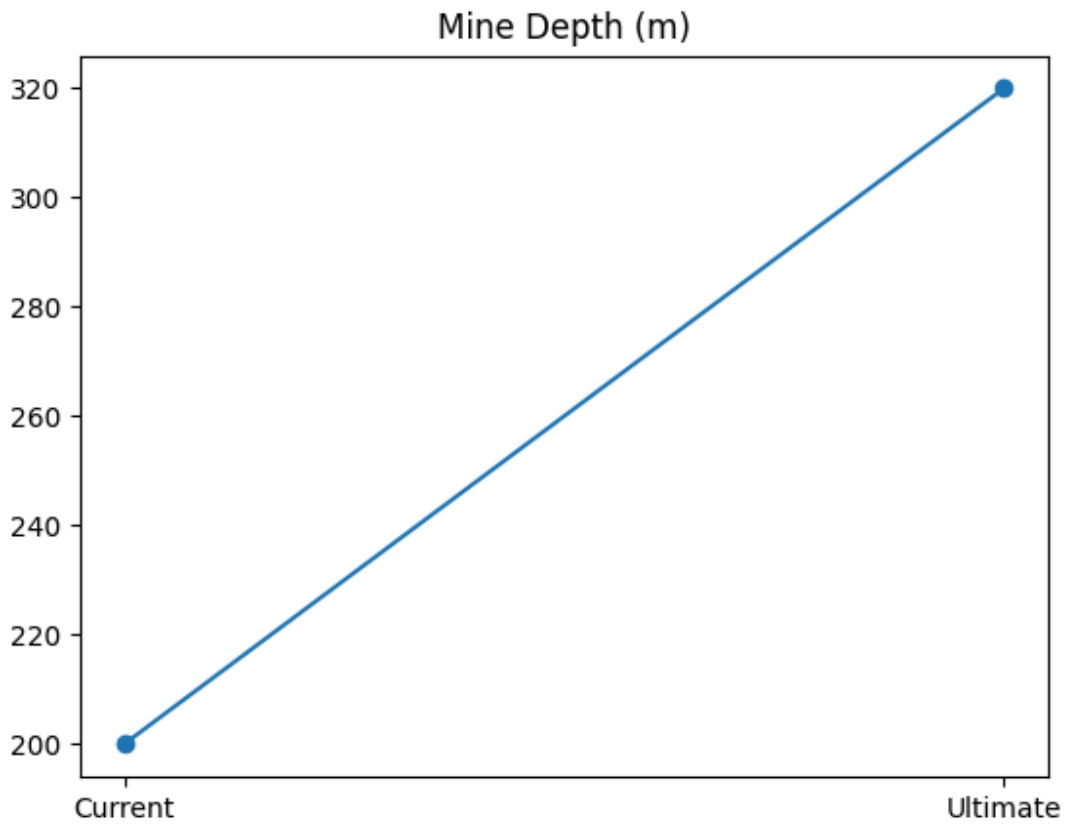
This chart shows the proportion of coal grades (G8–G11) produced at the mine.
Indicates the quality and calorific value of coal.
Significance: Important for pricing, blending, and suitability for thermal power plants like those of NTPC Limited.

📊 6. Bench Heights (m)



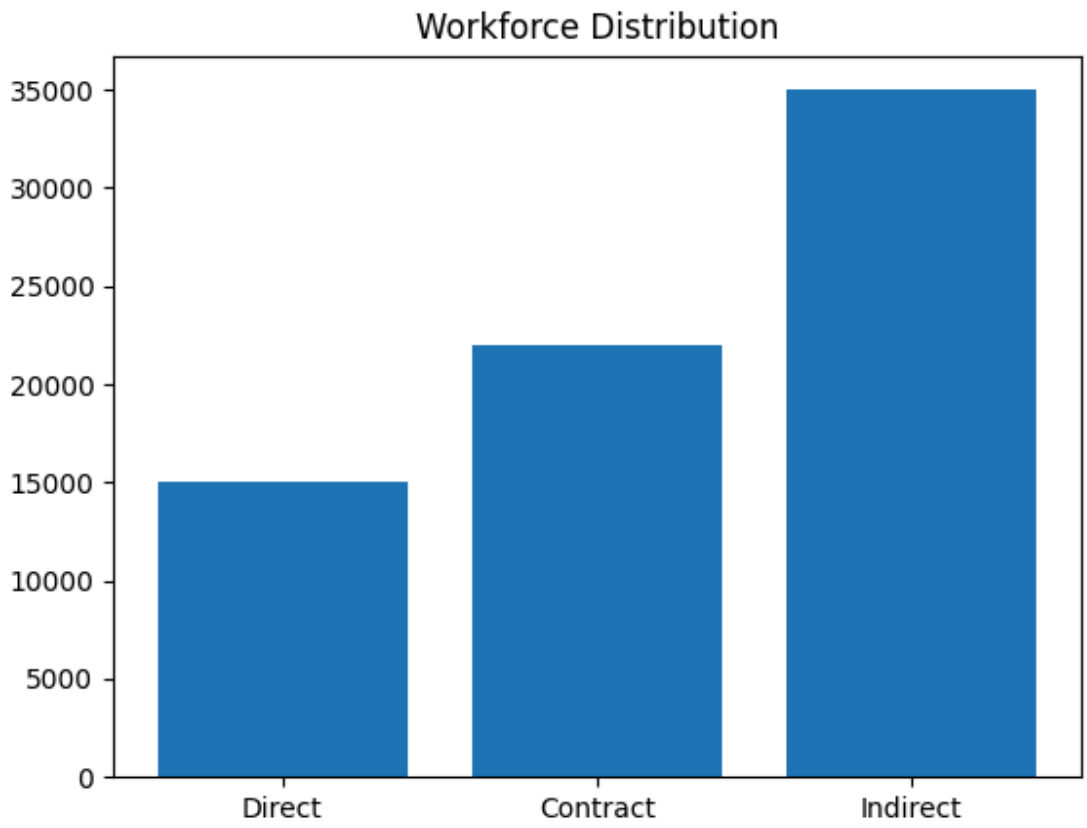
Represents the height of benches used in coal and overburden extraction.
Different heights are used based on equipment and safety requirements.
Significance: Essential for slope stability, safety, and efficient excavation.

📊 7. Mine Depth Progression



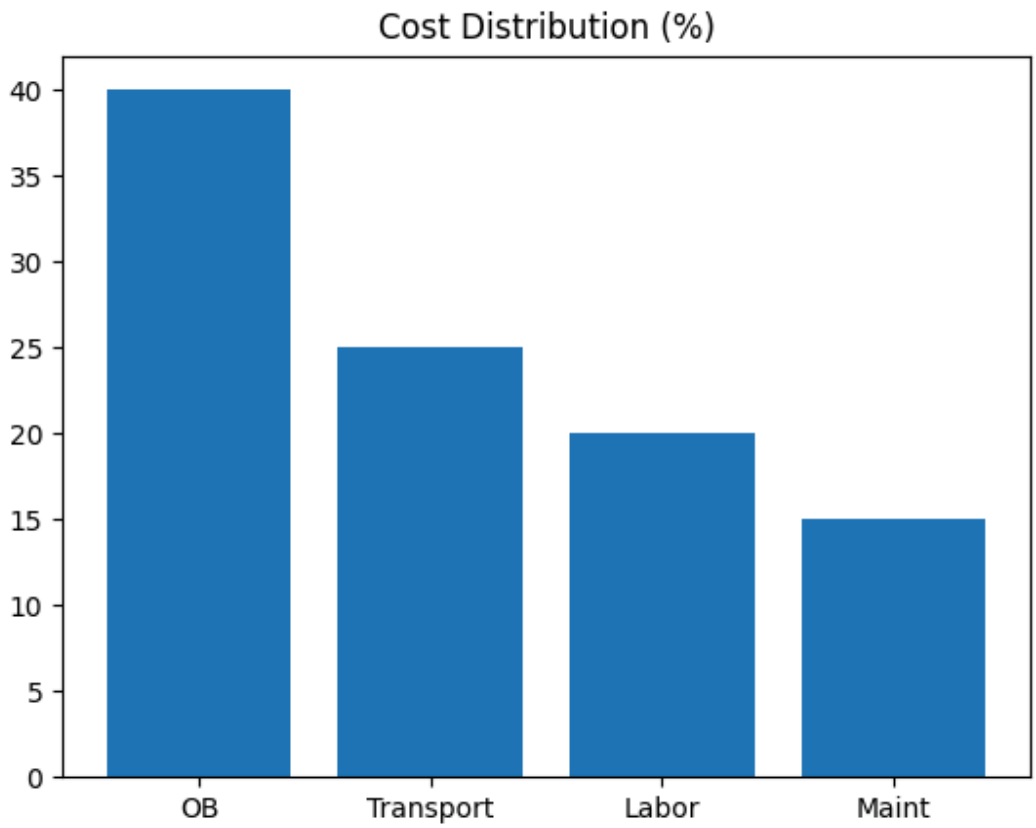
Shows current vs ultimate depth of the mine.
Indicates expansion and deepening of mining operations.
Significance: Important for long-term mine planning, slope design, and groundwater management.

📊 8. Workforce Distribution



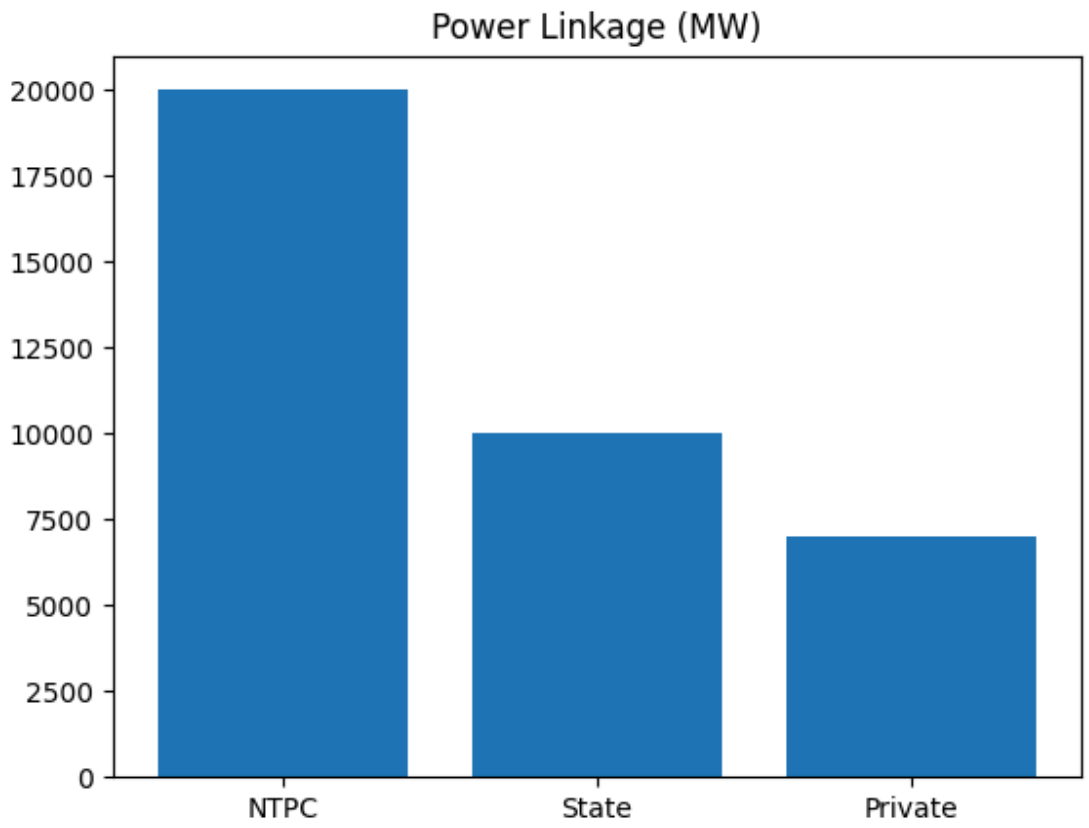
Displays distribution of manpower into direct, contract, and indirect categories.
Highlights employment generation by mining operations.
Significance: Useful for manpower planning and understanding socio-economic impact.

📊 9. Cost Distribution (%)



Shows how total mining cost is divided among OB removal, transport, labor, and maintenance. OB removal usually forms the largest share.
Significance: Helps in cost control, budgeting, and improving operational efficiency.

📊 10. Power Linkage Capacity (MW)



Represents coal supply linkage to different power sectors.
Jayant OCP supports major power generation units.
Significance: Demonstrates the mine's role in ensuring energy security and supporting industrial demand.